React Installations:-
->install node.js
->install react app
->creating first react application

Node js installation

What is node js?

Node.js is a free, open-source tool that lets you run JavaScript outside the browser. Because of this, you can use JavaScript to build many kinds of applications like websites, servers, APIs, and more.

Node.js uses the **V8 JavaScript engine** (the same engine used by Google Chrome), which makes it very fast and efficient.

A Node.js application runs as a **single process** instead of creating a new thread for every request. It handles multiple requests using **asynchronous (non-blocking) operations**, so the program does not stop while waiting for tasks like database access, file reading, or network responses.

When Node.js performs tasks such as reading files, fetching data from a database, or receiving network data, it does **not block the program**. Instead, it continues running other tasks and comes back to finish the operation once the response is ready. This helps Node.js use system resources efficiently and handle many users at the same time.

How to run?

To run this code, first save it in a file named `server.js`.
Then open your terminal and run:

```
node server.js
```

If the code uses the `.mjs` format, save it as `server.mjs` and run:

```
node server.mjs
```

How this Node.js server code works?

1. **Importing the http module**
   The code starts by including Node.js's built-in `http` **module**.

Node.js provides a powerful standard library, and it has excellent support for networking.

2. **Creating the server**
   The `http.createServer()` method is used to create a new HTTP server.
   This method returns a server object.

3. **Listening on a port**
   The server is set to listen on a specific **port number** and **host name**.
   Once the server starts successfully, a callback function runs and prints a message saying the server is running.

4. **Handling requests**
   Every time a client (like a browser) sends a request, the server triggers a request event.
   This event gives two important objects:

   - **request (`req`)** – contains information sent by the client (URL, headers, data, etc.)

   - **response (`res`)** – used to send data back to the client

5. **Using the request object**
   In this simple example, the request object is not used.
   However, it can be used to read request headers, query parameters, or request body data.

6. **Sending the response**
   The response object is used to send data back to the client.

   - `res.statusCode = 200;`
     This sets the HTTP status code to **200**, meaning the request was successful.

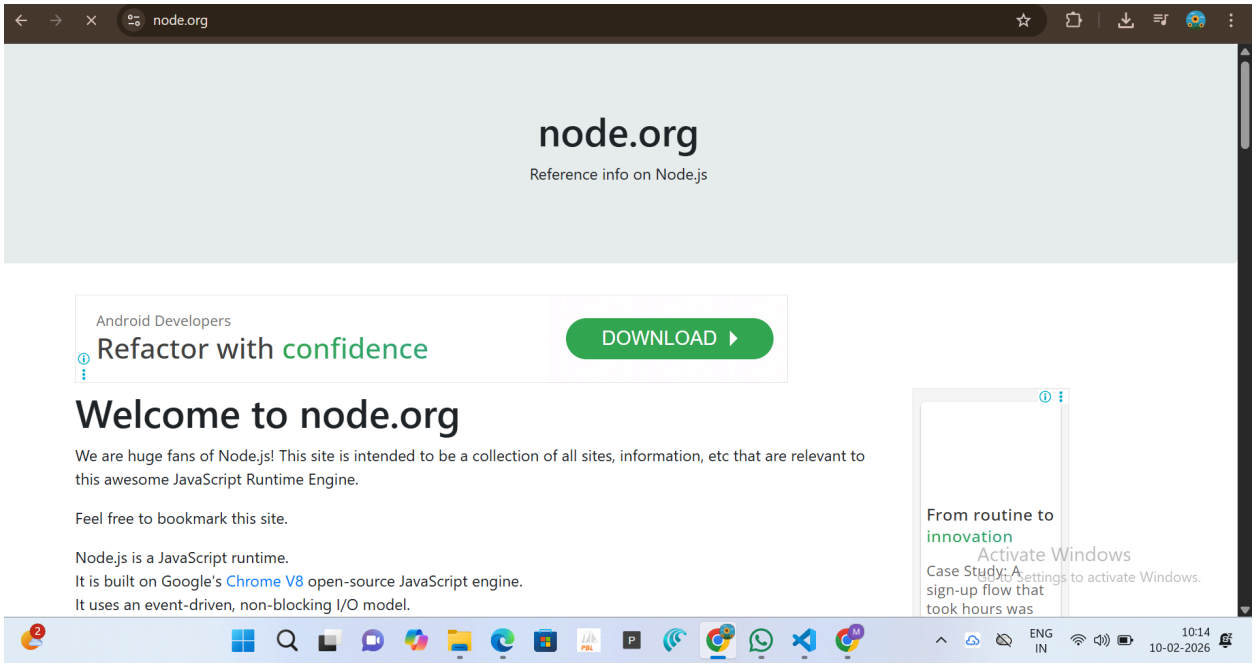
   - `res.setHeader('Content-Type', 'text/plain');`
     This tells the browser that the response is plain text.

   - `res.end('Hello World\n');`
     This sends the response content to the client and closes the response.

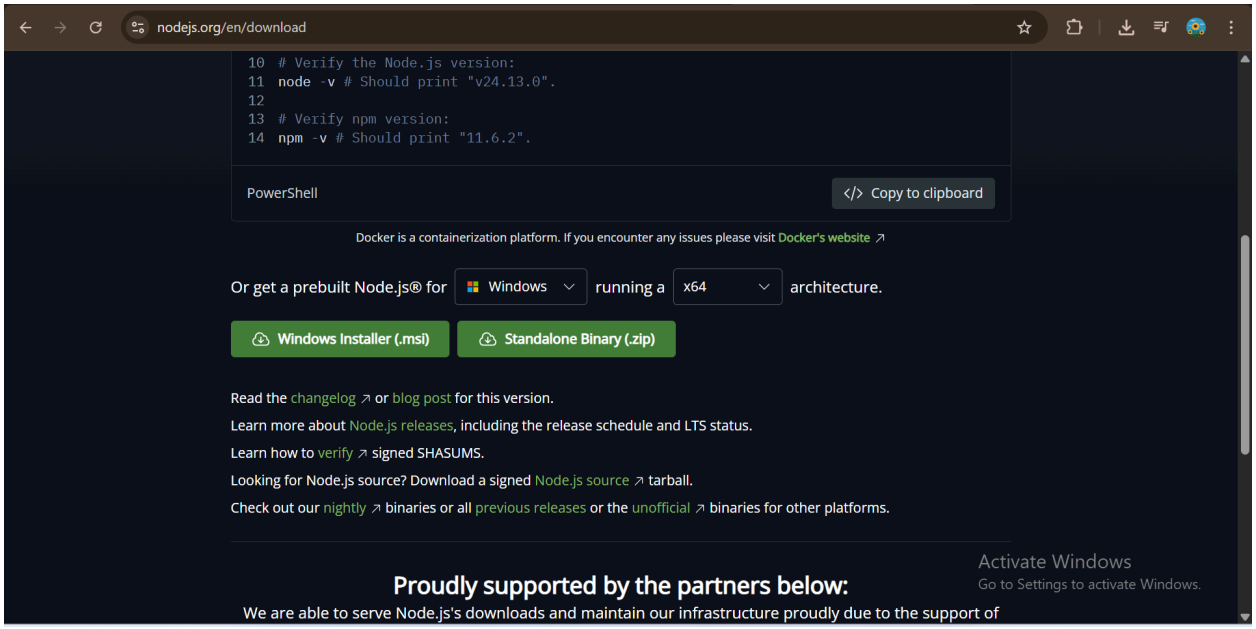
7. **Installing Node.js**
   If Node.js is not already installed, you need to download and install it before running this code.

Installation:-

Step1:-go to node.org website



Step2:- Choose the one that is adaptable for your system and click on download



Step3:- accept the terms and condition sand install .To check its correctly installed and check its version use command `node -v`

```
Command Prompt
Microsoft Windows [Version 10.0.22621.4317]
(c) Microsoft Corporation. All rights reserved.

C:\Users\neelamma>node -v
v22.18.0

C:\Users\neelamma>
```

What is npm?
npm is a tool used to install, manage, and share JavaScript packages (libraries) for Node.js projects.

To check type `npm`

```
C:\Users\neelamma>npm
npm <command>

Usage:

npm install        install all the dependencies in your project
npm install <foo>  add the <foo> dependency to your project
npm test           run this project's tests
npm run <foo>      run the script named <foo>
npm <command> -h   quick help on <command>
npm -l             display usage info for all commands
npm help <term>    search for help on <term> (in a browser)
npm help npm       more involved overview (in a browser)

All commands:

    access, adduser, audit, bugs, cache, ci, completion,
    config, dedupe, deprecate, diff, dist-tag, docs, doctor,
    edit, exec, explain, explore, find-dupes, fund, get, help,
    help-search, hook, init, install, install-ci-test,
    install-test, link, ll, login, logout, ls, org, outdated,
    owner, pack, ping, pkg, prefix, profile, prune, publish,
    query, rebuild, repo, restart, root, run-script, sbom,
    search, set, shrinkwrap, star, stars, start, stop, team,
    test, token, uninstall, unpublish, unstar, update, version,
    view, whoami

Specify configs in the ini-formatted file:
    C:\Users\neelamma\.npmrc
or on the command line via: npm <command> --key=value

More configuration info: npm help config
Configuration fields: npm help 7 config
```

Activate Windov
Go to Settings to activ

What is react?

React is an open-source JavaScript library developed by Facebook for building user interfaces. It uses a component-based architecture and a virtual DOM to efficiently update and render UI components, making web applications fast and scalable.

How to build a react project?

Step 1: Install prerequisites

Make sure these are installed:

Install Node.js

1.Download from: https://nodejs.org

2.Install LTS version

3.Check installation:

```
node -v
```

```
npm -v
```

Step 2: Create a React project

To go to your desired folder to create a project,use command

```
cd Desktop\edutainer\react learning\Day9AndDay10"
```

Vite is faster and easier than Create React App

Run this in terminal:

```
npm create vite@latest my-react-app
```

You'll see questions:

- **Framework** → `React`
- **Variant** → `JavaScript` (or JSX)

Step 3: Go into the project folder

```
cd my-react-app
```

Step 4: Install dependencies

```
npm install
```

Step 5: Start the development server

```
npm run dev
```

Open browser and go to:

```
http://localhost:5173
```

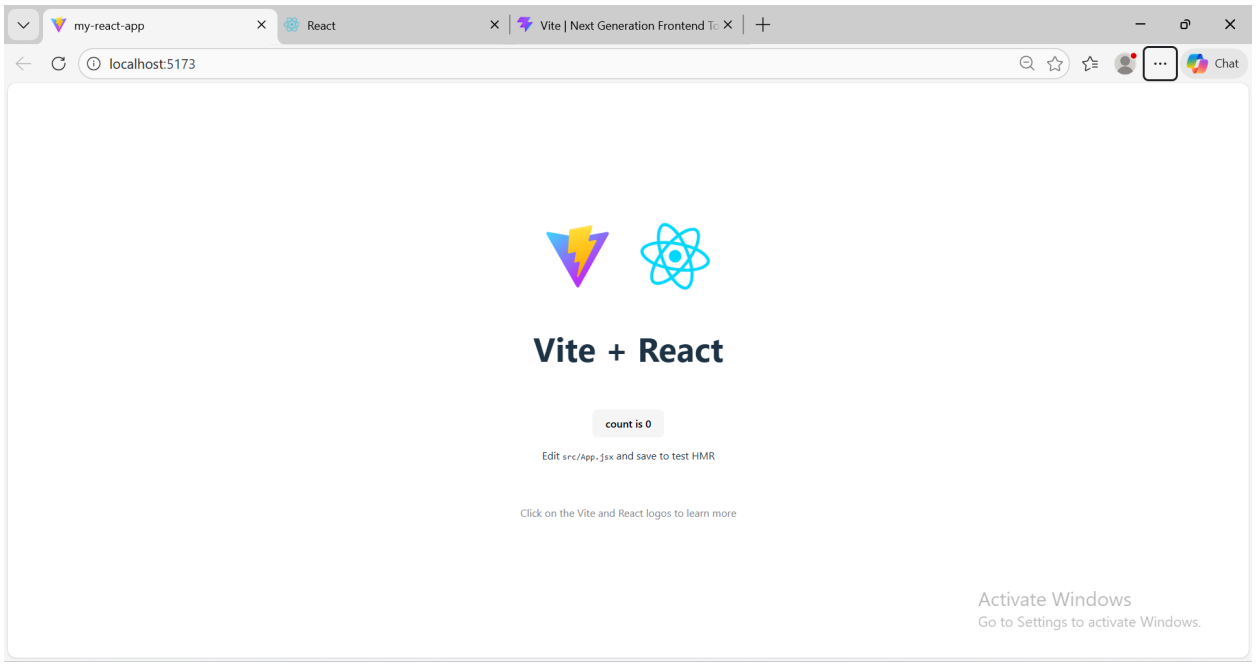
Your React app is running!

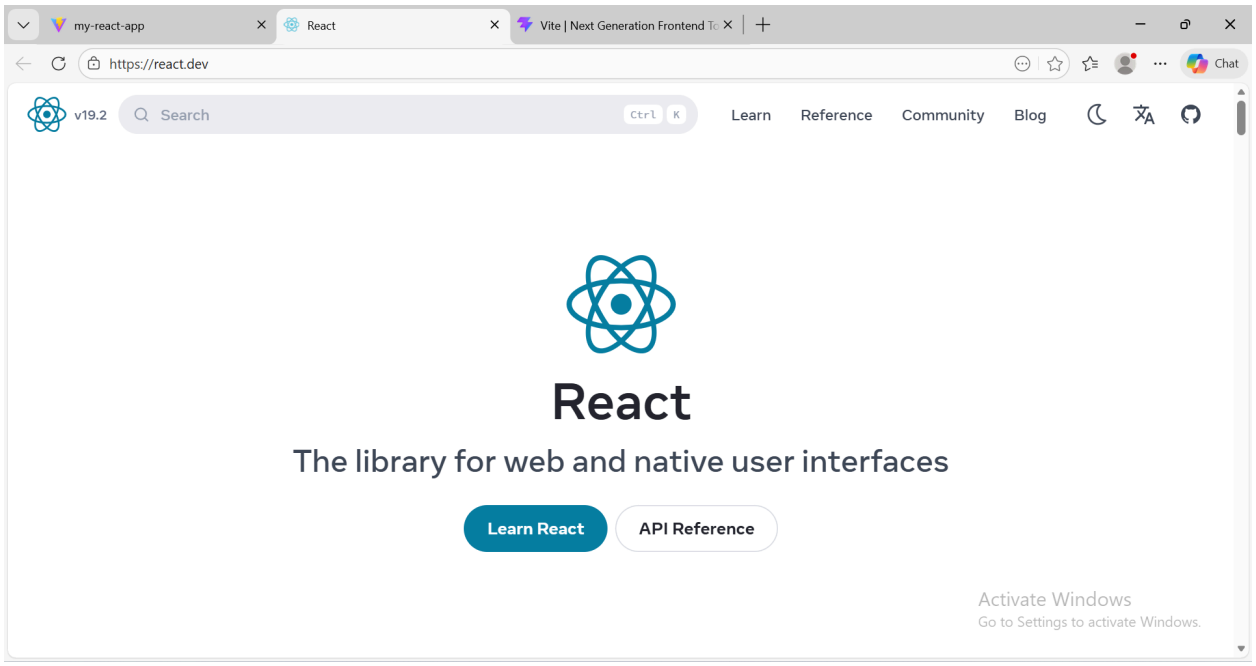
```
Microsoft Windows [Version 10.0.22621.4317]
(c) Microsoft Corporation. All rights reserved.

C:\Users\neelamma>cd Desktop\edutainer\react learning\Day9AndDay10

C:\Users\neelamma\Desktop\edutainer\react learning\Day9AndDay10>npm create vite@latest my-react-app
Need to install the following packages:
create-vite@8.3.0
Ok to proceed? (y) y


> npx
> create-vite my-react-app

|
o  Select a framework:
|  React
|
o  Select a variant:
|  JavaScript + SWC
|
o  Use Vite 8 beta (Experimental)?:
|  Yes
|
o  Install with npm and start now?
|  Yes
|
o  Scaffolding project in C:\Users\neelamma\Desktop\edutainer\react learning\Day9AndDay10\my-react-app...
|
o  Installing dependencies with npm...

added 156 packages, and audited 157 packages in 5m
```

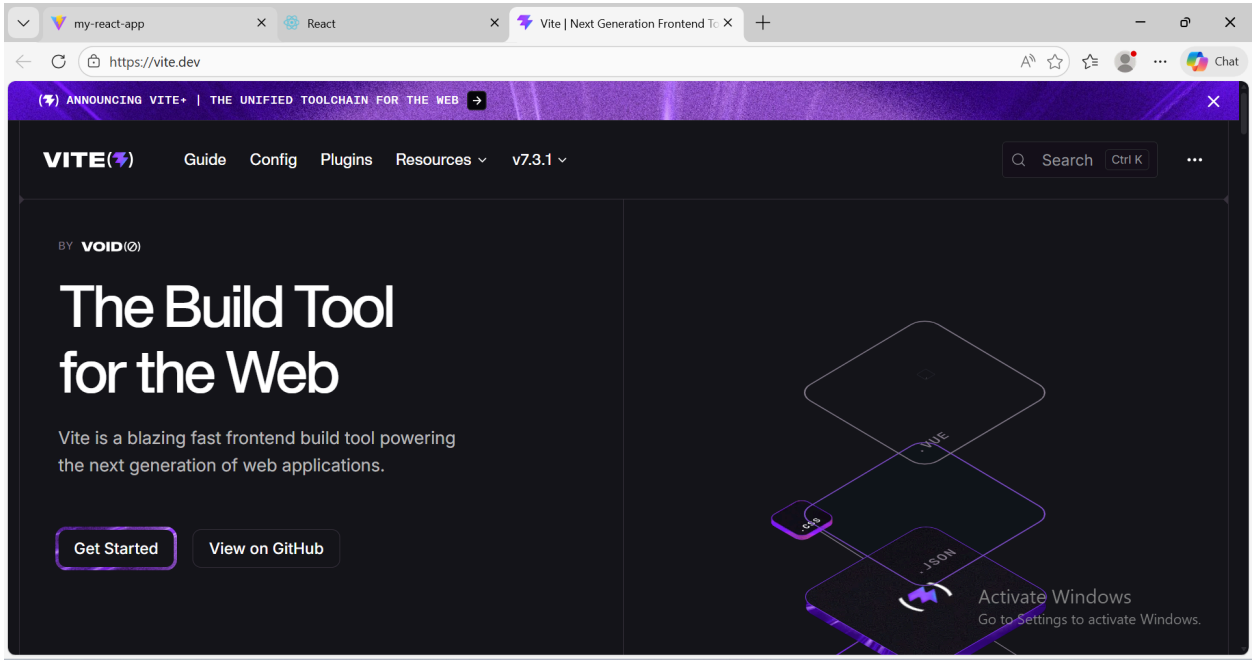
Activate Windows
Go to Settings to activate Windows.

```
C:\Windows\system32\cmd.e:

|  JavaScript + SWC
|
o  Use Vite 8 beta (Experimental)?:
|  Yes
|
o  Install with npm and start now?
|  Yes
|
o  Scaffolding project in C:\Users\neelamma\Desktop\edutainer\react learning\Day9AndDay10\my-react-app...
|
o  Installing dependencies with npm...

added 156 packages, and audited 157 packages in 5m

37 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
|
o  Starting dev server...

> my-react-app@0.0.0 dev
> vite

[vite:react-swc] We recommend switching to `@vitejs/plugin-react` for improved performance as no swc plugins are used. More informati
on at https://vite.dev/rolldown

  VITE v8.0.0-beta.13  ready in 648 ms

  ➜  Local:   http://localhost:5173/
  ➜  Network: use --host to expose
  ➜  press h + enter to show help
```

Activate Windows
Go to Settings to activate Windows.


my-react-app
React
Vite | Next Generation Frontend To
localhost:5173
Chat
Vite + React
count is 0
Edit src/App.jsx and save to test HMR
Click on the Vite and React logos to learn more
Activate Windows
Go to Settings to activate Windows.


my-react-app
React
Vite | Next Generation Frontend To
https://react.dev
Chat
v19.2
Search
Ctrl
K
Learn
Reference
Community
Blog
React
The library for web and native user interfaces
Learn React
API Reference
Activate Windows
Go to Settings to activate Windows.


my-react-app
React
Vite | Next Generation Frontend To
https://vite.dev
Chat
ANNOUNCING VITE+ | THE UNIFIED TOOLCHAIN FOR THE WEB
VITE
Guide
Config
Plugins
Resources
v7.3.1
Search
Ctrl K
BY VOID
The Build Tool for the Web
Vite is a blazing fast frontend build tool powering the next generation of web applications.
Get Started
View on GitHub
.VUE
.CSS
.JSON
Activate Windows
Go to Settings to activate Windows.